

Structured Probability: From Joints to Bayesian Networks

Factorization, conditional independence, and compact models

Learning objectives

By the end of this lesson, you should be able to:

- 01 ——— Compute joint, marginal, conditional, and posterior probabilities.
- 02 ——— Distinguish independence from conditional independence.
- 03 ——— Quantify the full-joint parameter explosion.
- 04 ——— Read variables, edges, parents, and CPTs in a DAG.
- 05 ——— Compute a full assignment from local factors.
- 06 ——— Identify graph-implied independencies informally.

Structure is a compression claim

A Bayesian network is not merely a diagram; its missing edges assert conditional independencies that permit factorization.

- Nodes are random variables.
- Directed edges identify local parent sets.
- Each CPT conditions a node on its parents.

Factorization

The graph and CPTs together define one joint distribution.

Probability identities remain the foundation

$$P(X|Y)=P(X,Y)/P(Y); P(X|Y)=P(Y|X)P(X)/P(Y)$$

- A joint assigns probability to complete combinations.
- Marginalization sums out unwanted variables.
- Conditioning renormalizes within observed evidence.
- Bayes' rule reverses a conditional using a prior.

ENGINEERING IMPLICATION

Every posterior calculation must account for both likelihood and base rate.

Independence changes after conditioning

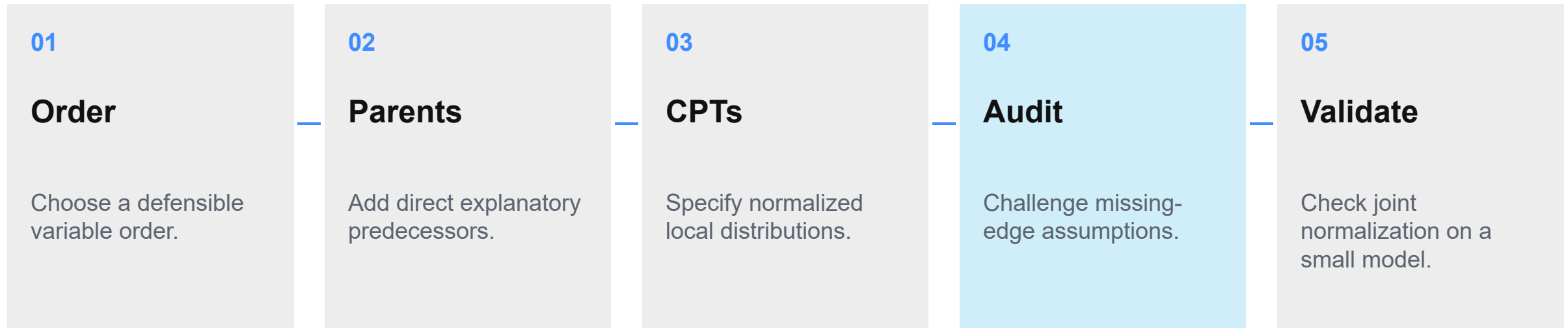
Marginal independence

- $P(X,Y)=P(X)P(Y)$.
- Observing X gives no information about Y .
- Can fail in a mixed population.
- Written X independent Y .

Conditional independence

- $P(X,Y|Z)=P(X|Z)P(Y|Z)$.
- Z explains the association.
- Can hold within each group.
- Written X independent Y given Z .

Build a Bayesian network deliberately



Every omitted parent should correspond to an independence assumption you can defend.

Probability of a complete assignment

```
function JOINT_PROBABILITY(network, assignment):  
    p = 1  
    for variable in topological_order(network):  
        value = assignment[variable]  
        parent_values = assignment[parents(variable)]  
        p = p * CPT(variable, value, parent_values)  
    return p
```

Implementation contract

- Require a DAG.
- Use a complete assignment.
- Select the correct CPT row.
- Check probabilities in $[0,1]$.

Bayes rule: a positive test is not a diagnosis

SCENARIO

Disease prevalence is 1%, sensitivity is 90%, and false-positive rate is 5%.

- $P(D \text{ and } +) = 0.01(0.90) = 0.009$.
- $P(\text{not } D \text{ and } +) = 0.99(0.05) = 0.0495$.
- $P(+) = 0.0585$.
- $P(D|+) = 0.009/0.0585$ about 0.154.

Despite 90% sensitivity, a positive result implies only about a 15.4% posterior because false positives act on a much larger group.

Representations scale differently

Full joint

Every assignment

- 2^n rows for binary variables.
- $2^n - 1$ free parameters.
- No structural assumptions.
- Direct but infeasible.

Sparse Bayes net

Local CPTs

- Parents stay small.
- Parameters grow locally.
- Independence is explicit.
- Queries need inference.

Data reality

Estimate reliably

- Rare cells lack samples.
- CPTs pool by parent context.
- Structure can be wrong.
- Validation remains essential.

Checkpoint: base-rate reasoning

CHECKPOINT

If prevalence falls while sensitivity and false-positive rate stay fixed, what happens to $P(\text{disease}|\text{positive})$?

01

Write numerator and denominator.

02

Identify which population dominates false positives.

03

State the direction, not only intuition.

Common graph patterns imply different evidence flow

Chain and fork

- $X \rightarrow Z \rightarrow Y$ or $X \leftarrow Z \rightarrow Y$.
- X and Y may be dependent.
- Conditioning on Z blocks the path.
- Z explains transmission or cause.

Collider

- $X \rightarrow Z \leftarrow Y$.
- X and Y can be marginally independent.
- Conditioning on Z opens dependence.
- Explaining away appears.

Bayesian-network chain rule

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

- The order is topological.
- Root nodes use marginal priors.
- Each nonroot factor selects a CPT row.
- Multiplication yields one full-assignment probability.

ENGINEERING IMPLICATION

A sparse parent structure replaces one exponential table with many small tables.

Worked joint from a four-node network

SCENARIO

Rain R has $P(R)=0.2$; Sprinkler S has $P(S|R)=0.1$; WetGrass W has $P(W|R,S)=0.99$; Slippery L has $P(L|W)=0.7$.

- Choose assignment R,S,W,L all true.
- Select factors 0.2, 0.1, 0.99, and 0.7.
- Multiply in topological order.
- $0.2(0.1)(0.99)(0.7)=0.01386$.

The graph converts a complete-world probability into four auditable local lookups.

Graph defects have recognizable consequences

Cycle

Invalid BN

- No topological order.
- Circular local semantics.
- Factorization undefined.
- Reject at construction.

Missing edge

False independence

- Model too simple.
- Posterior can be biased.
- Residual dependence remains.
- Audit with domain knowledge.

Extra edge

Lost compression

- More CPT rows.
- More data required.
- Still can be valid.
- May overfit estimates.

Checkpoint: parameter count

CHECKPOINT

How many free parameters does a full joint over 20 binary variables require, and why is one subtracted?

01

Count assignments.

02

Apply normalization.

03

Compare with 20 independent Bernoulli roots.

Validate both numbers and semantics

A syntactically valid DAG can still encode the wrong causal or conditional story.

Audit

Reviewers need normalized CPTs, graph assumptions, and tests on known conditional relationships.

- Check every CPT row sums to one.
- Generate samples and compare empirical marginals with analytic values.
- Document that edges represent probabilistic dependence, not automatically causation.

Key takeaways

- 01 ——— Bayes' rule combines likelihood and prior.
- 02 ——— Conditional independence is context-specific.
- 03 ——— A Bayesian network factors a joint through parent sets.
- 04 ——— Sparse structure makes representation compact but creates an inference task.

Next: Exact and approximate inference turn the factorized model into posterior answers.

Check your understanding

Q1

Compute a diagnostic posterior.

Q2

Distinguish a fork from a collider.

Q3

Count full-joint parameters.

Q4

Verify one CPT row.

Q5

Compute a complete assignment.